

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 1– Concept Mapping**

|                         |                                                                                                                                                                                                                                                                                                                           |                      |                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------|
| <b>Course Code:</b>     | CSA0312                                                                                                                                                                                                                                                                                                                   | <b>Course Title:</b> | Data Structures                     |
| <b>CO Assessed:</b>     | CO1 – Imitation (BL3, BL4)                                                                                                                                                                                                                                                                                                | <b>Assessment:</b>   | Assessment Tool 1 – Concept Mapping |
| <b>Weightage:</b>       | 10% of CIA                                                                                                                                                                                                                                                                                                                | <b>Total Marks:</b>  | 100                                 |
| <b>CO1 Statement:</b>   | Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis. |                      |                                     |
| <b>Student Name:</b>    | priya dharshini M                                                                                                                                                                                                                                                                                                         | <b>Reg. No.:</b>     | 192521051                           |
| <b>Section / Batch:</b> | 2025                                                                                                                                                                                                                                                                                                                      | <b>Date:</b>         | 26-07-2026                          |

**Code of Conduct**

*I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:

**Instructions**

- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

**Rubrics**

| Questions | Problem Understanding | Method Selection | Solution Process | Accuracy of Calculation | Interpretation of Results | Total Marks |
|-----------|-----------------------|------------------|------------------|-------------------------|---------------------------|-------------|
| Q1        |                       |                  |                  |                         |                           |             |
| Q2        |                       |                  |                  |                         |                           |             |

## Annexure A – Concept Mapping

Questions:

EACH QUESTIONS CARRIES : 10 MARKS

Duration :ONE HOUR

Q17 A logistics company processes delivery requests across multiple cities. Construct a concept map linking Algorithm → Input Size → Time Complexity → Space Complexity → Scalability → Efficiency. Analyze how algorithm selection impacts system scalability and justify the best approach for handling large datasets.

